Experiment 38: Temperature Variability Analysis

Aim: To calculate mean, standard deviation and temperature range for different cities.

Algorithm:

1. Import Pandas.
2. Create temperature data for cities.
3. Calculate mean temperature.
4. Calculate standard deviation and temperature range.
5. Find the city with the highest range and lowest standard deviation.

Code:

```
import pandas as pd

df = pd.DataFrame({
   "Chennai": [30, 32, 35, 33, 31],
   "Delhi": [20, 25, 30, 35, 40],
   "Mumbai": [28, 29, 30, 29, 28]
})

print("Mean:\n", df.mean())
print("Standard Deviation:\n", df.std())

ranges = df.max() - df.min()

print("Temperature Range:\n", ranges)
print("Highest Range City:", ranges.idxmax())
print("Most Consistent City:", df.std().idxmin())
```

Input:

Chennai: 30, 32, 35, 33, 31

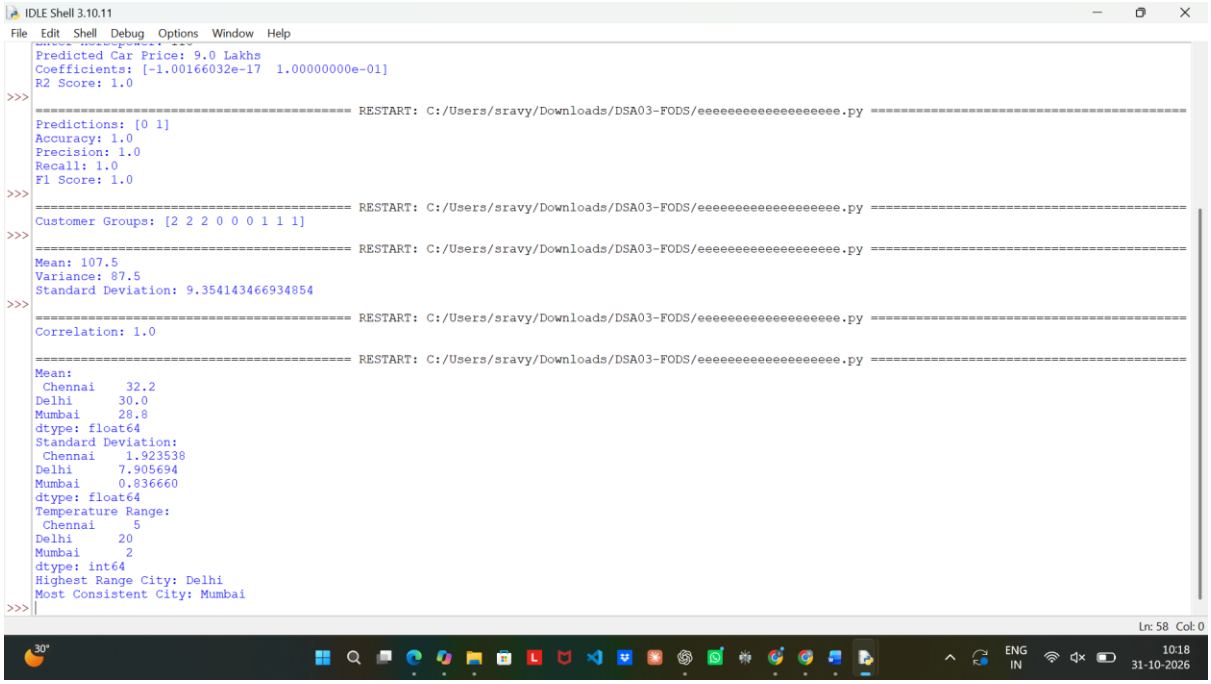
Delhi: 20, 25, 30, 35, 40

Mumbai: 28, 29, 30, 29, 28

Output:

Highest Range City: Delhi

Most Consistent City: Mumbai



Result: Temperature variability and consistency were calculated for three cities.